

Challenge: Figure Eight

Points: 150 • Estimated Time: 90 Minutes

Challenge Scope

The robot must autonomously drive in a smooth figure-eight pattern around two cones spaced one meter apart.

Instructions & Engineering Hints

- Welcome to the **Figure Eight Challenge**! Your mission is to program your robot to autonomously drive in a continuous, smooth figure-eight pattern around two cones spaced exactly one meter apart.
- Start by planning your path: a figure-eight requires two contrasting turns (one clockwise, one counter-clockwise) that intersect at a central crossing point.
- **Architectural Hint (Steering & Kinematics):** Consider how your robot executes turns. Differential drive robots achieve smooth curves by varying the relative speeds of the left and right motors, rather than relying on sharp pivot turns.
- **Sensor & Navigation Tips:** While you can use dead reckoning (timed motor rotations and delays), think about how external feedback sensors like a gyro, compass, or camera could help correct drift to ensure your loops remain symmetrical over multiple runs.
- **Iterative Testing:** Test your code one loop at a time. Perfect your first circle before tackling the transition into the crossing point and the second opposing circle.